

Take Home Summary

Links

- Dashboard: <https://cascade-regional-health-dashboard.streamlit.app/>
- Github Repo: https://github.com/ntran22/Truveta_Take_Home_Assessment

Dashboard Summary

Cascade Regional Health's population health analysis reveals a significant burden of chronic disease across the patient population, with the most prevalent conditions including:

- Anemia/Blood Disorders (10.6%)
- Diabetes (9.4%)
- Asthma (8.7%)
- Depression (8.1%)
- COPD/Chronic Respiratory Disease (8.0%)
- Myocardial Infarction (7.9%)
- Cardio-Kidney Disease (7.6%)

These findings suggest opportunities to strengthen preventive care, chronic disease management, and longitudinal patient engagement programs to improve outcomes and reduce downstream healthcare utilization.

While the majority of patients (**81.6%**) fall outside of high-risk utilization categories, the dashboard identified a concentrated subset of patients who may disproportionately impact healthcare resources:

- 8.4% are Super Utilizers Only
- 5.0% are High-Cost Patients Only
- 5.0% are High-Cost & Super Utilizer Overlap

Collectively, these segments account for 18.4% of the patient population and represent a priority opportunity for value-based care intervention. Recommended next steps include implementing targeted care coordination and chronic disease management programs for the High-Cost & Super Utilizer Overlap population, developing utilization reduction strategies for Super Utilizers, and proactively managing the highly prevalent chronic conditions identified within the population. By focusing resources on these high-risk patient segments and chronic disease cohorts, Cascade Regional Health can improve patient outcomes, optimize resource utilization, and advance its value-based care objectives.

Condition Prevalence by Condition Group



Overall Encounter Patient Population: High-Cost and Super Utilizer Segments



Path to Production

To productize this dashboard for deployment across health systems, additional clinical, operational, and governance capabilities would be required. The current solution uses encounter, diagnosis, patient demographic, and billing data to provide proxy measures for population health, utilization, and care management opportunities. However, many value-based care and quality metrics cannot be fully supported with the provided files. For example, formal HEDIS, CMS Stars, and quality reporting programs would require additional data elements such as member enrollment and attribution, payer information, laboratory results, medication adherence data, pharmacy claims, provider attribution, referral data, social determinants of health, and longitudinal outcome measures. Clinical validation with healthcare stakeholders would also be necessary to ensure condition groupings, risk stratification logic, PMPY calculations, and care management definitions align with operational and quality improvement workflows.

Before external deployment or scaling to additional health systems, the solution would require enhancements across engineering, user experience, governance, and performance domains. Recommended improvements include automated data ingestion pipelines, data quality monitoring, role-based access controls, audit logging, HIPAA-compliant security controls, and scalable cloud infrastructure capable of supporting larger patient populations and near real-time refreshes. The dashboard should also undergo user-centered design validation with clinical, operational, and executive stakeholders to refine workflows, improve usability, and ensure actionable decision-making. Finally, benchmarking capabilities, configurable quality measures, customer-specific attribution methodologies, and standardized metric definitions would be needed to support broad adoption across diverse healthcare organizations while maintaining consistency, trust, and regulatory compliance.